

CHUAN ZHANG

Victoria, BC, Canada | charleszhang92@gmail.com | github.com/zhangchuan92910

PROFILE:

Mathematics M.Sc. student combining algorithm research, quantum open-source development and prior product engineering experience. Seeking quantum algorithms research roles and internships.

EDUCATION:

University of Victoria, Victoria, Canada August 2027 (expected)
M.Sc. in Mathematics | GPA: 9.00/9.00

- Coursework: Quantum Algorithms and Software Engineering (100%); Computational Complexity (99%); Discrete Optimization (95%).

Simon Fraser University, Burnaby, Canada 2022
B.Sc. in Computing Science, Minor in Mathematics

QUANTUM OPEN SOURCE:

IonQ Core Python | unitaryHACK 2026 June - July 2026
Open Source Contributor | Bounty recipient | PR #75

- Implemented four pure-Python helpers to convert quantum measurement probabilities into shot counts, bitstrings, marginal distributions and Z-parity expectation values.
- Added unit tests for rounding, qubit ordering and input-validation edge cases; the results module reached 100% branch coverage.

PROFESSIONAL HISTORY:

Roboterra Inc., Beijing, China February 2016 - September 2017
Product Engineer

- Integrated drivers for new robot accessories into the backend cloud compiler, extending support for additional hardware.
- Expanded educational robot accessories and developed interactive courses teaching programming and physics to teenagers.

RESEARCH EXPERIENCE:

University of Victoria, Victoria, Canada
Master's Research in Graph Algorithms | Supervisor: Jing Huang

- Designed a polynomial-time recognition algorithm and proved structural characterizations for a class of strongly chordal directed graphs.
- Constructed and proved vertex orderings for further directed-graph families, extending the constructions using the supervisor's structural framework.

SKILLS:

Programming: Python, C, C++ | Tools: Git, LaTeX

Quantum computing: PennyLane, Qiskit, measurement result post-processing

Methods: Algorithm design, computational complexity, graph theory, discrete optimization